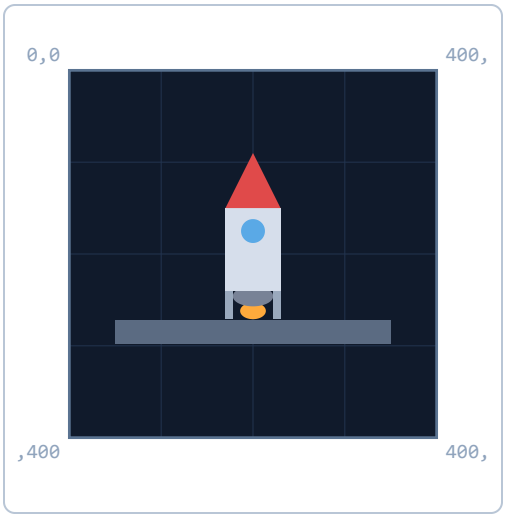
Build the Rocket



What you are making

A 400 × 400 sketch with eight shapes, using every drawing call you know: `rect`, `triangle`, `ellipse` and `circle`. Every measurement is given — this is about getting the numbers into the right places, in the right order.

```
function setup() {
  createCanvas(400, 400);
  // this line matters - see below
  rectMode(CENTER);
}
```

> Every measurement in the table is a **center point**, so `rectMode(CENTER)` must be set in `setup()` before any of it works.

The measurements

Draw them **in this order**.

SHAPE	CALL	CENTER	SIZE	COLOR
Landing pad	`rect`	`200, 285`	`300 × 26`	`91, 107, 130`
Exhaust plume	`ellipse`	`200, 262`	`28 × 18`	`255, 170, 60`
Engine bell	`ellipse`	`200, 246`	`44 × 22`	`120, 130, 150`
Body	`rect`	`200, 195`	`60 × 90`	`214, 222, 235`
Nose cone	`triangle`	`200, 90  230, 150  170, 150`		`224, 74, 74`
Window	`circle`	`200, 175`	`diameter 26`	`90, 169, 230`
Left leg	`rect`	`174, 255`	`8 × 30`	`154, 169, 189`
Right leg	`rect`	`226, 255`	`8 × 30`	`154, 169, 189`

Background: `background(16, 26, 43)` · use `noStroke()`.

> **The order is part of the assignment.** Later shapes cover earlier ones. Draw the engine bell before the plume and the plume vanishes behind it — move the lines, not the shapes.

> **The round shapes are measured differently.** `ellipse` and `circle` are always positioned from their center, whatever `rectMode` says. `circle` takes a *diameter*, not a width and a height.

Turning it in

1. Name your sketch `u1l03-rocket` and **save it**.
2. Go to `File → Share`.

3. Copy the **View-Only** link. Not Edit, not Embed.
4. Paste it as a **comment** on the Classroom post for this assignment.
5. Open your own link in a new tab and check it actually works.

> **A sketch you never saved shares an empty link.** Save first, then share. Step 5 takes ten seconds and catches this.

If you finish early

- Add stars to the sky — there is only one place in `draw()` where they end up behind everything else.
- Add a second stage, a hatch, or an exhaust flame.
- Try changing `createCanvas(400, 400)` to `600, 600`. What breaks, and why?

Stuck? Check the console first — it is underneath the code in the p5 editor and it tells you which line is unhappy. Then the reference sheet. Then ask. Not finished by the bell: share the link anyway and keep working tonight; the same link shows your latest saved work.